

06 Vector Space

6.2 Linear Combination

Lines, Planes, and Linear Interpolation in RGB Color Space — the Affine Geometry Behind Five Linear Combinations



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Linear combination: a weighted sum of a set of vectors and their corresponding scalars (coefficients), producing a new vector

3

$\mathbf{a}_1 + k_2\mathbf{a}_2 + k_3\mathbf{a}_3$: a plane through the endpoint of $\mathbf{a}_1$, parallel to $\mathbf{a}_2$ and $\mathbf{a}_3$

5

$k\mathbf{1} + (1-k)\mathbf{a}$: the line through the endpoints of $\mathbf{1}$ and $\mathbf{a}$, lightening a color toward white

2

$\mathbf{a}_1 + k\mathbf{a}_2$: a line through the endpoint of $\mathbf{a}_1$, parallel to $\mathbf{a}_2$

4

$k\mathbf{a}_1 + (1-k)\mathbf{a}_2$: the line through the endpoints of $\mathbf{a}_1$ and $\mathbf{a}_2$, i.e. linear interpolation

6

$k_1\mathbf{a}_1 + k_2\mathbf{a}_2 + (1-k_1-k_2)\mathbf{a}_3$: the plane through the endpoints of $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$

Five Linear Combinations, Five Geometric Shapes

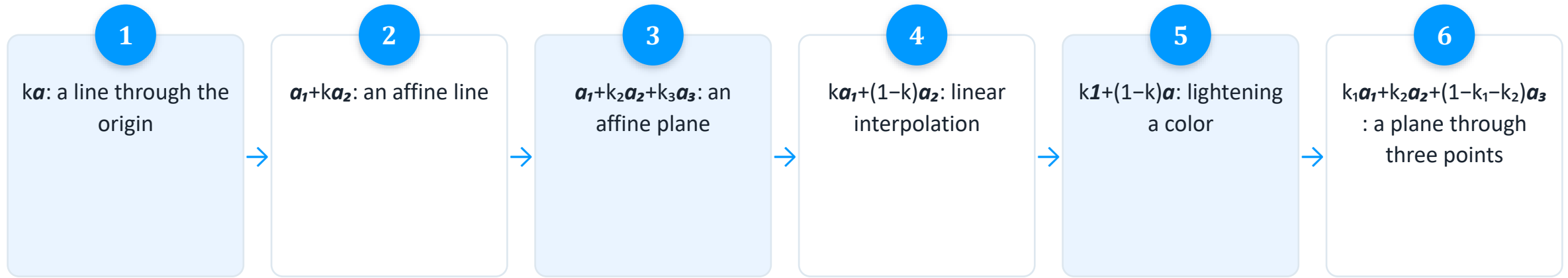
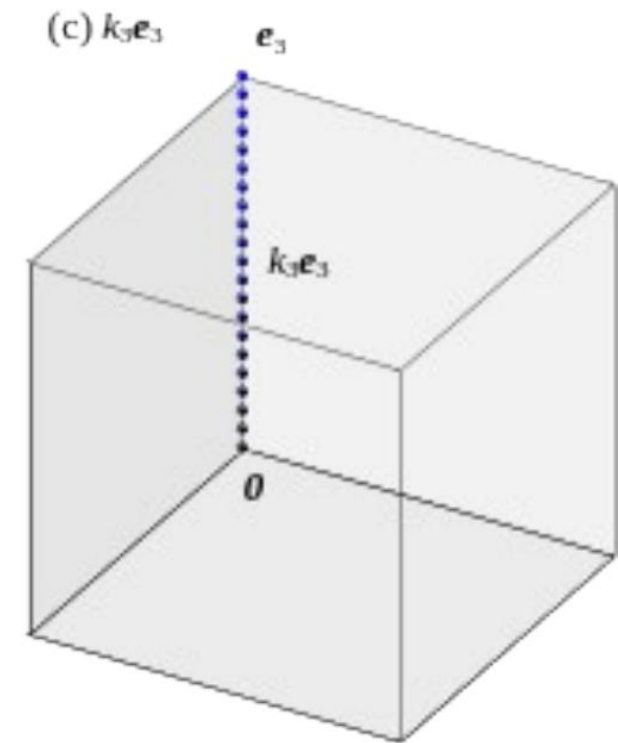
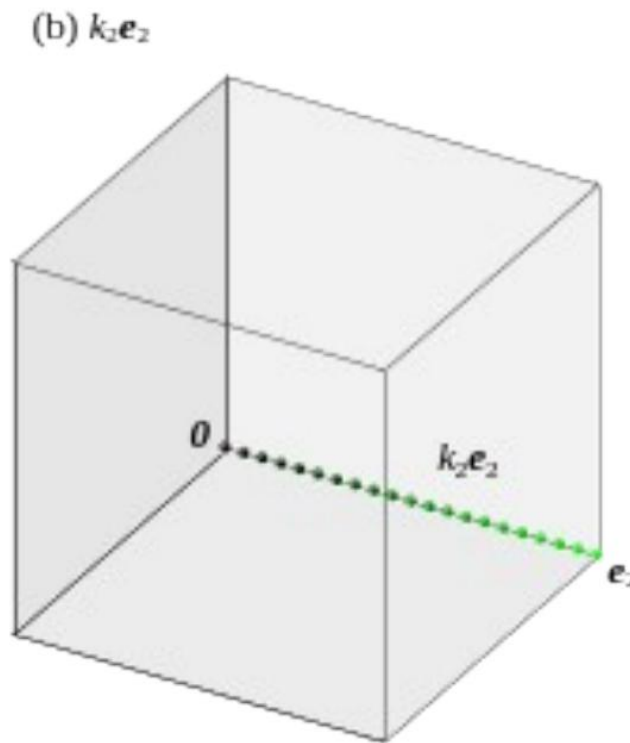
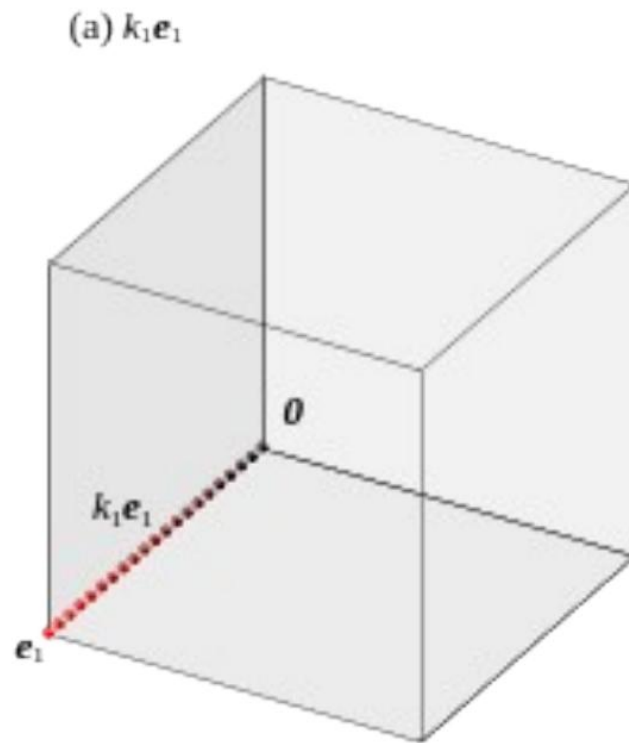


Figure 1 · ke_1, ke_2, ke_3 Each Span a One-Dimensional Space

- (a)(b)(c) are k_1e_1, k_2e_2, k_3e_3 — each spans a line through the origin, $\text{span}(e_i)$



$$A_1 + KA_2$$

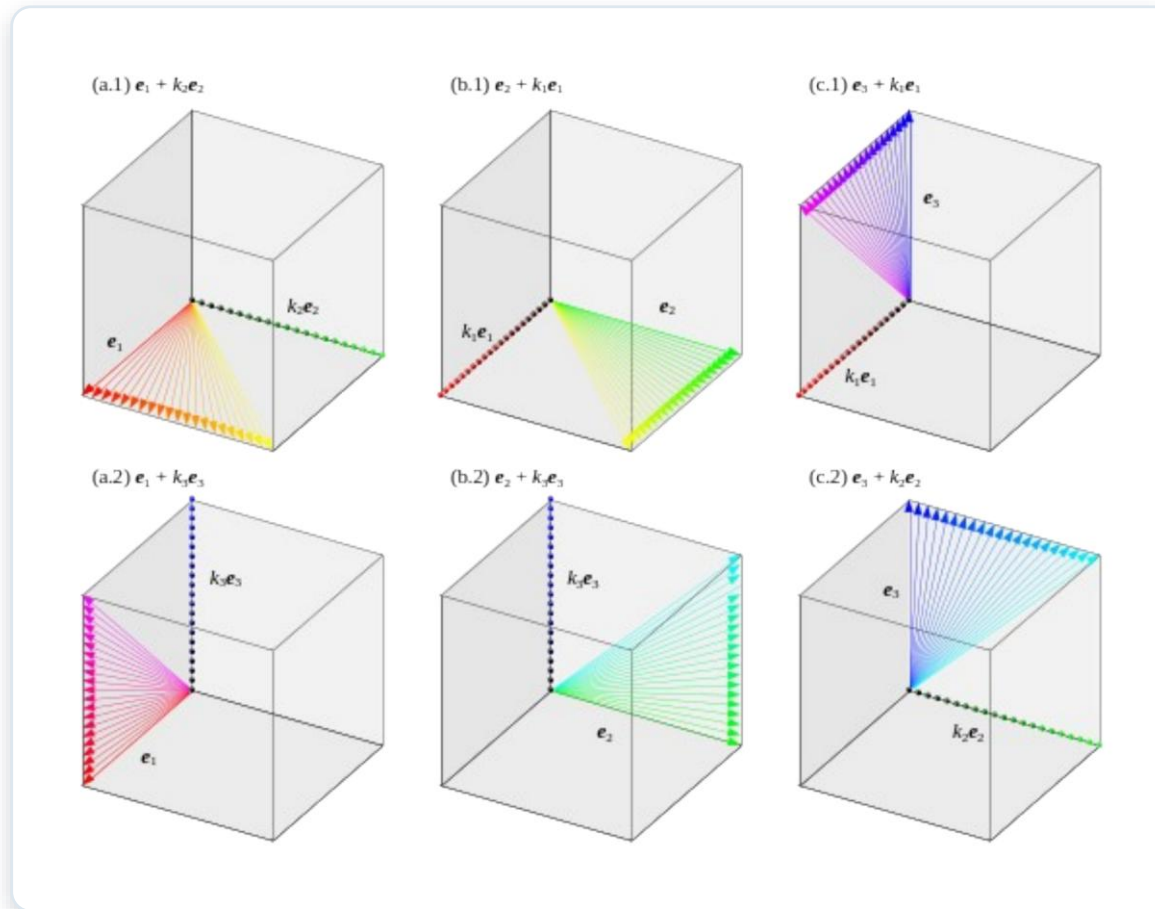
A Line Through the Endpoint of a_1 , Parallel to a_2

$$a_1 + ka_2$$

- Given a_2 as an n-dimensional column vector and k any real number, ka_2 spans a 1-D space $\text{span}(a_2)$ that passes through the origin 0
- $a_1 + ka_2$ shifts that line $\text{span}(a_2)$ by a_1 , producing an affine space — no longer through the origin

Figure 2 · $a_1 + ka_2$ in RGB Color Space

- In (a.1): $e_1 + k_2e_2$ shifts from red to yellow — this line excludes pure black, i.e. it misses the origin



$$A_1 + K_2A_2 + K_3A_3$$

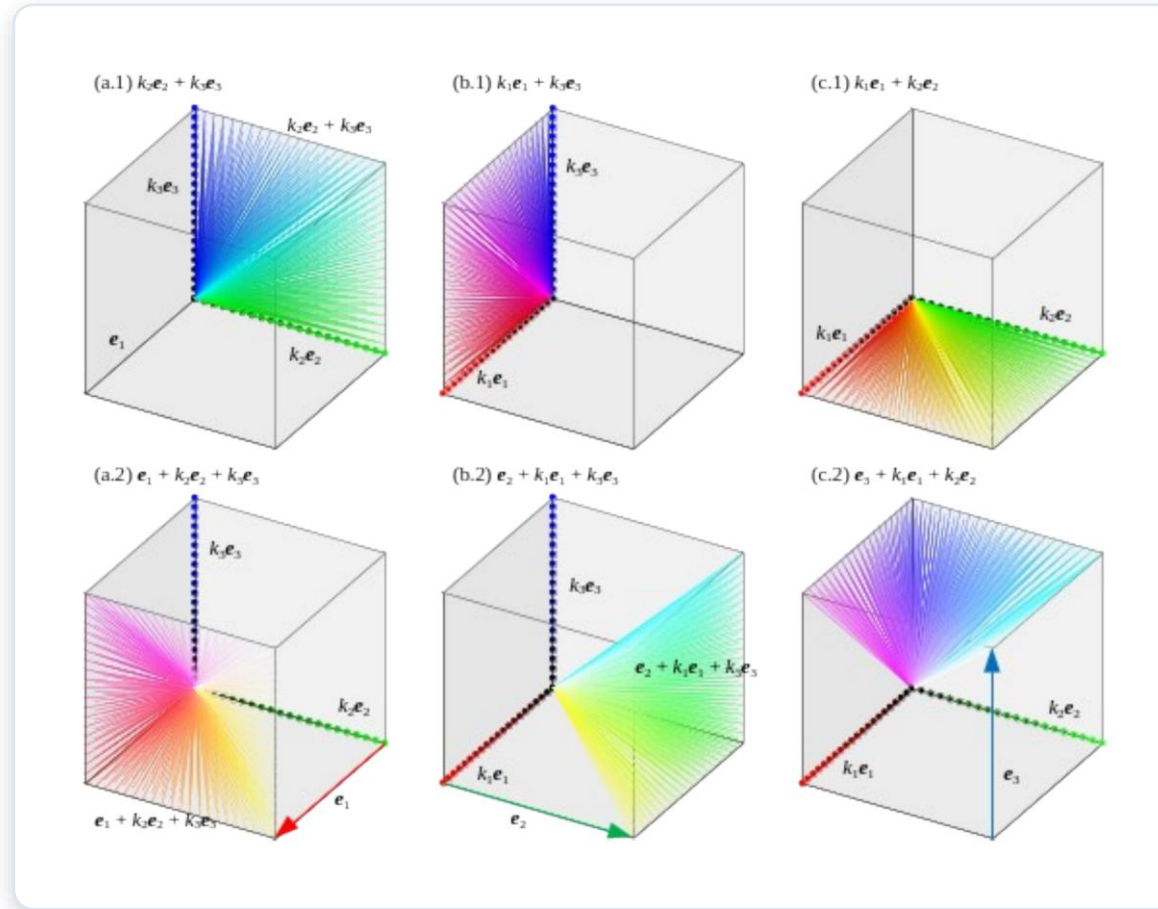
A Plane Through the Endpoint of a_1 , Parallel to a_2 and a_3

$$a_1 + k_2a_2 + k_3a_3$$

- For arbitrary real k_2, k_3 , the term $k_2a_2 + k_3a_3$ spans a 2-D plane $\text{span}(a_2, a_3)$ through the origin 0
- $a_1 + (k_2a_2 + k_3a_3)$ shifts that plane by a_1 , producing an affine space

Figure 3 · $a_1 + k_2a_2 + k_3a_3$ in RGB Color Space

- In (a.1): $k_2e_2 + k_3e_3$ spans a "blue-green" plane through the origin; shifting by e_1 gives (a.2), no longer through the origin



$$KA_1 + (1 - K)A_2$$

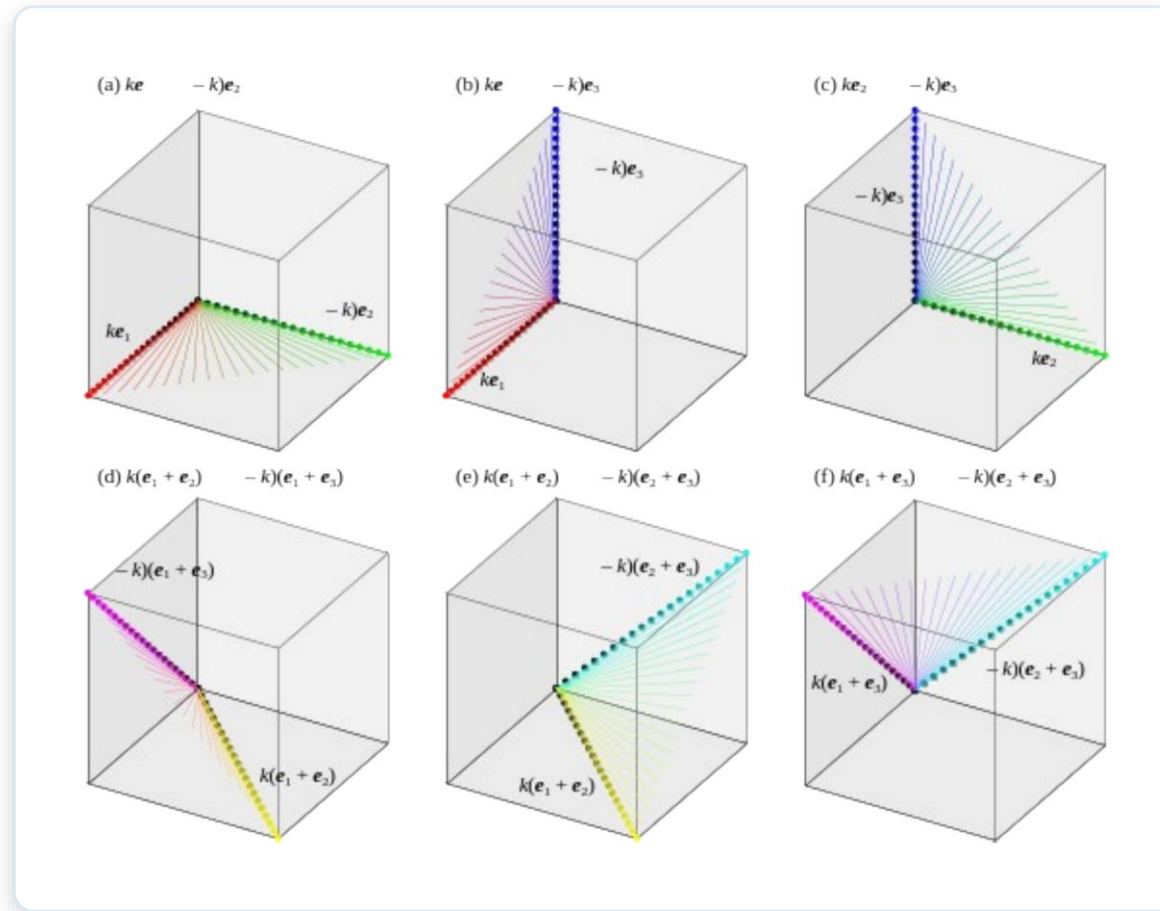
Linear Interpolation: the Line Through a_1 and a_2

$$ka_1 + (1 - k)a_2$$

- For arbitrary real k , $ka_1 + (1-k)a_2$ traces the line through the endpoints of a_1 and a_2 — linear interpolation
- Geometrically, linear interpolation draws a straight line between two points and locates intermediate points along it

Figure 4 · $ka_1 + (1 - k)a_2$ in RGB Color Space

- (a) shows interpolation between red and green — the segment is exactly the line between the endpoints of the red and green vectors



$$K\mathbf{1} + (1 - K)A$$

Linear Interpolation That Lightens a Color

$$k\mathbf{1} + (1 - k)\mathbf{a}$$

- The all-ones column vector $\mathbf{1}$ represents white; $\mathbf{a}$ can be any color vector; the scalar k ranges over $[0, 1]$
- The closer k is to 1, the closer the color is to white (lighter); the closer k is to 0, the closer it is to the original color $\mathbf{a}$

Figure 5 · $k\mathbf{1} + (1 - k)\mathbf{a}$ in RGB Color Space

- For $k\mathbf{1} + (1 - k)\mathbf{e}_i$: the color is red when $k = 0$, and white when $k = 1$

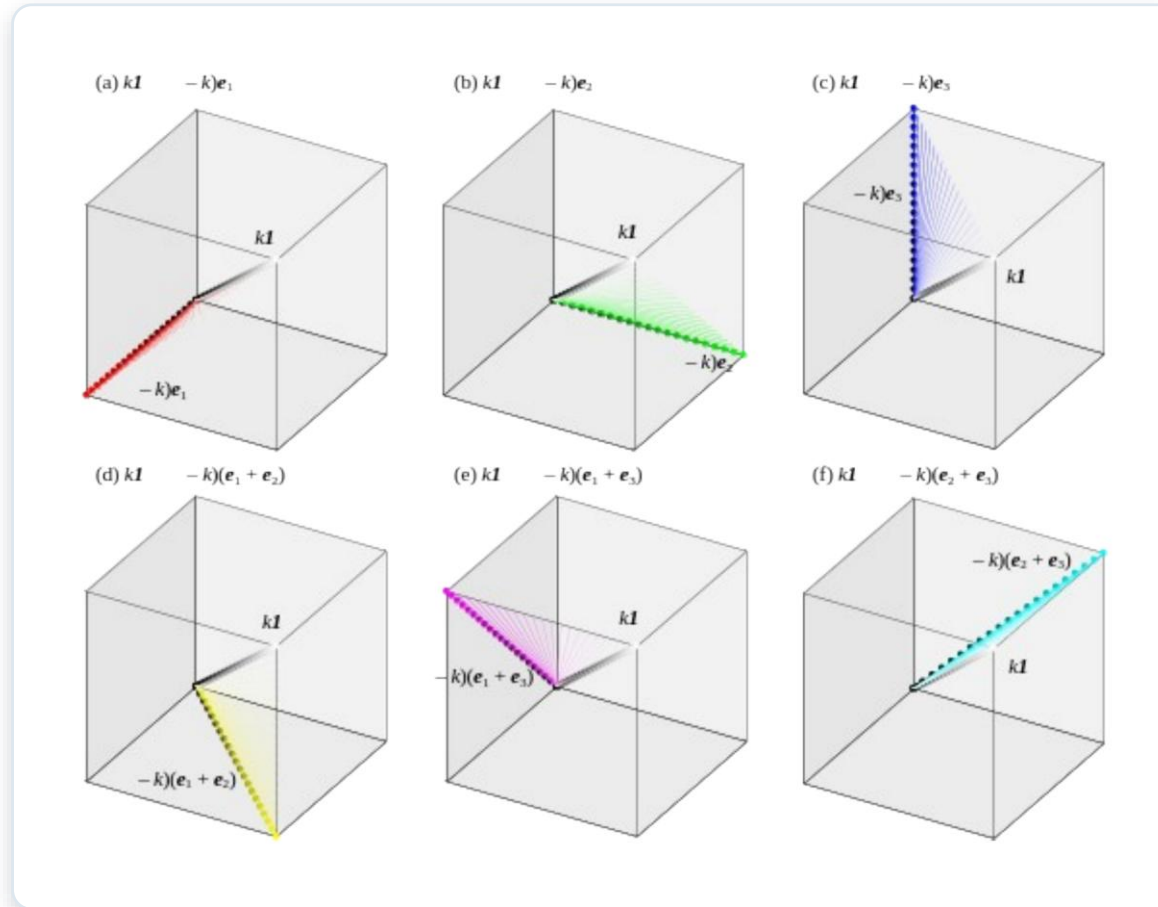
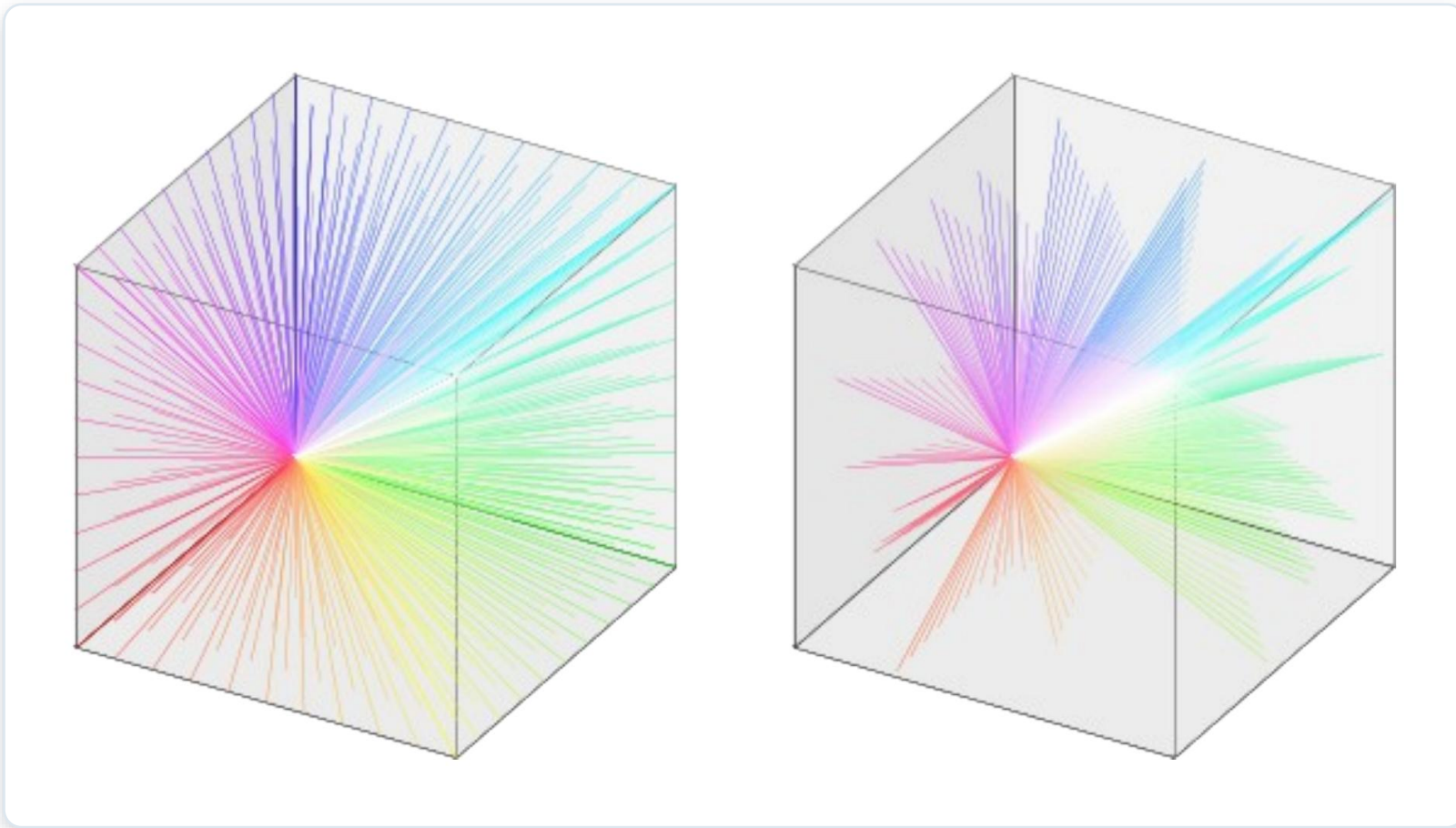


Figure 6 · Creative Coding for Color Lightening in RGB Space

- Both examples apply $k\mathbf{1} + (1-k)\mathbf{a}$ to many color vectors at once, letting them converge toward white in a radial gradient



$$K_1A_1 + K_2A_2 + (1 - K_1 - K_2)A_3$$

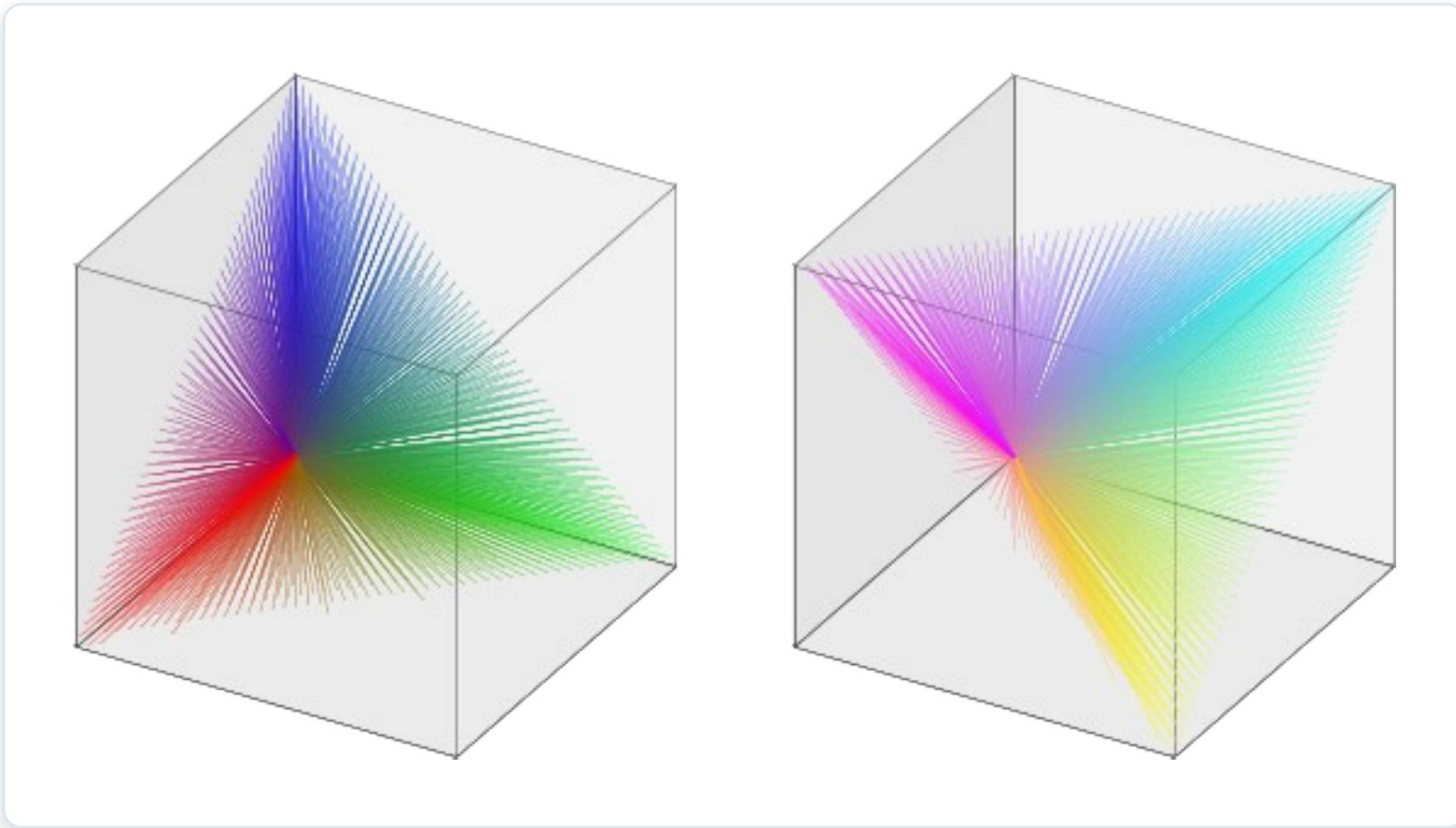
A Plane Through the Endpoints of a_1, a_2, a_3

$$k_1a_1 + k_2a_2 + (1 - k_1 - k_2)a_3$$

- This linear combination spans a plane passing through the endpoints of the three vectors a_1, a_2, a_3
- The three coefficients always sum to 1 — the hallmark of an affine combination; requiring nonnegative coefficients as well makes it a convex combination

Figure 7 · A Plane Spanned in RGB Color Space

- (a) is the plane fixed by the red, green, blue vector endpoints; (b) is the plane fixed by yellow, magenta, cyan



Five Linear Combinations, Five Geometric Shapes

Line $\rightarrow$ Plane $\rightarrow$ Linear Interpolation $\rightarrow$ Color Lightening $\rightarrow$ Three-Point Plane

- $\mathbf{a}_1 + k\mathbf{a}_2$ and $\mathbf{a}_1 + k_2\mathbf{a}_2 + k_3\mathbf{a}_3$ shift $\text{span}()$ to get affine spaces — a line and a plane, respectively
- $k\mathbf{a}_1 + (1-k)\mathbf{a}_2$ and $k\mathbf{1} + (1-k)\mathbf{a}$ are two applications of linear interpolation: connecting two points, and lightening a color
- $k_1\mathbf{a}_1 + k_2\mathbf{a}_2 + (1-k_1-k_2)\mathbf{a}_3$ has coefficients summing to 1 — an affine combination spanning the plane through three points

Glossary for This Section

Linear Combination 线性组合 —— a weighted sum of vectors and their corresponding scalars

Linear Interpolation 线性插值 —— $ka_1 + (1-k)a_2$, the line connecting two points

Convex Combination 凸组合 —— an affine combination with nonnegative coefficients

Scalar 标量 —— the weighting coefficient k in a linear combination

Through the Origin 过原点 —— the test for a vector space: whether $\mathbf{0}$ belongs to it

Affine Space 仿射空间 —— a vector space after translation; no longer contains the origin

Affine Combination 仿射组合 —— a linear combination whose coefficients always sum to 1

Span 张成 —— the set of all linear combinations of a set of vectors

All-Ones Vector 全1向量 $\mathbf{1}$ —— a column vector of all 1s; represents white in RGB

RGB Color Space 本节贯穿始终的几何直观载体，用于理解线性组合的仿射几何

Further Reading

Further Reading · Affine and Convex Combinations

Coefficients summing to 1 make an affine combination; nonnegative coefficients summing to 1 make a convex combination — the basis of barycentric coordinates in computer graphics

Further Reading · Creative Coding and Color Gradients

The principle $k\mathbf{1} + (1-k)\mathbf{a}$ generates radial color gradients — a common technique in data visualization and generative art

Review · Vector Spaces and Span

Every affine space in this section builds on the vector space and `span()` concepts from §6.1

Open-Source Repository

Full code and image assets: <https://github.com/Visualize-ML>

Exercises Q1 – Q7

Q1 Using NumPy, construct $\mathbf{a}_1=[1,0]$, $\mathbf{a}_2=[0,1]$; for $k=-2,-1,0,1,2$ compute $\mathbf{a}_1+k\mathbf{a}_2$ and verify the points are collinear.

Q3 Use NumPy to verify that $\text{span}(\mathbf{a}_2)$ passes through the origin: substitute $k=0$ into $k\mathbf{a}_2$ and check whether the result is the zero vector $\mathbf{0}$.

Q5 Using Matplotlib's 3D tools, sample a grid of (k_2, k_3) values and plot the plane spanned by $\mathbf{a}_1+k_2\mathbf{a}_2+k_3\mathbf{a}_3$.

Q7 Implement a linear-interpolation function `lerp(a1, a2, k)` in NumPy that returns $k*a1+(1-k)*a2$; verify that at $k=0.5$ the result is the midpoint of $a1$ and $a2$.

Q2 Use Matplotlib to plot the line traced by $\mathbf{a}_1+k\mathbf{a}_2$ as k varies continuously over $[-3,3]$; mark the point at $k=0$, i.e. $\mathbf{a}_1$.

Q4 Construct 3-D vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ in NumPy; for several (k_2, k_3) pairs compute $\mathbf{a}_1+k_2\mathbf{a}_2+k_3\mathbf{a}_3$ and verify the points are coplanar.

Q6 Write a function `is_on_plane(p, a1, a2, a3)` that uses `numpy.linalg.lstsq()` to test whether point p lies on the affine plane spanned by $a1, a2, a3$.

Exercises Q8 – Q14

Q8 Use Matplotlib to visualize a color gradient bar for linear interpolation between red and green as k runs from 0 to 1.

Q10 Write a function `lighten(color, k)` implementing $k*1+(1-k)*color$, lightening any RGB color toward white as k runs from 0 to 1.

Q12 Following Figure 6, write code that batch-generates a lightening animation (k varying linearly from 0 to 1) for several color vectors at once.

Q14 Write a function `affine_combo(vectors, weights)` that checks whether weights sum to 1 and, if so, returns the corresponding affine combination.

Q9 Use NumPy to verify that $k*a1+(1-k)*a2$ and $(1-k)*a2+k*a1$ give identical results (commutativity).

Q11 Use Matplotlib to plot the color-gradient sequence produced by `lighten(color, k)` at $k=0,0.2,0.4,0.6,0.8,1$.

Q13 Use NumPy to verify that the three coefficients of $k1*a1+k2*a2+(1-k1-k2)*a3$ always sum to 1.

Exercises Q15 – Q20

Q15 Building on Q14, write a function `convex_combo(vectors, weights)` that additionally checks whether all weights are nonnegative.

Q17 Generate 100 random (k_1, k_2) pairs in NumPy, compute the corresponding $k_1 \cdot \mathbf{a}_1 + k_2 \cdot \mathbf{a}_2 + (1 - k_1 - k_2) \cdot \mathbf{a}_3$, and verify all the points lie on the same plane.

Q19 Use SymPy symbolic computation to verify that the distance from $\mathbf{a}_1 + k\mathbf{a}_2$ (with $\mathbf{a}_2$ nonzero) to $\mathbf{a}_1$ grows as $|k|$ increases.

Q16 Use Matplotlib's 3D tools to plot the plane from Figure 7(a) fixed by the red, green, blue vector endpoints, marking the three vertices.

Q18 Explain: when $k_1 \geq 0$, $k_2 \geq 0$, $k_1 + k_2 \leq 1$, what region of the plane does $k_1 \mathbf{a}_1 + k_2 \mathbf{a}_2 + (1 - k_1 - k_2) \mathbf{a}_3$ trace out? Visualize it with Matplotlib.

Q20 Comprehensive exercise: use ipywidgets sliders to adjust k_1 , k_2 , and display in real time the position of $k_1 \mathbf{a}_1 + k_2 \mathbf{a}_2 + (1 - k_1 - k_2) \mathbf{a}_3$ alongside a Figure-7-style plane rendering.

10 Prompts to Explore with AI

1 Explain why the geometric shape of $\mathbf{a}_1 + k\mathbf{a}_2$ is a full line rather than a ray — what values can k take?

2 Prove that in $k_1\mathbf{a}_1 + k_2\mathbf{a}_2 + (1 - k_1 - k_2)\mathbf{a}_3$ the three coefficients always sum to 1 — why is this called an "affine combination"?

3 Explain the difference between affine combinations and convex combinations, and how convex combinations are used in barycentric coordinate interpolation in computer graphics.

4 Demonstrate in Python how to linearly interpolate any RGB color toward a lighter shade using $k\mathbf{1} + (1 - k)\mathbf{a}$.

5 Explain why $\mathbf{a}_1 + k_2\mathbf{a}_2 + k_3\mathbf{a}_3$ spans an affine space rather than a vector space.

10 Prompts to Explore with AI

6 Design an animation script showing how, as k varies from 0 to 1, the point $k\mathbf{a}_1 + (1-k)\mathbf{a}_2$ moves from $\mathbf{a}_1$ to $\mathbf{a}_2$.

7 Explain the connection and the difference between linear interpolation and linear combination.

8 Give a real-world example of affine spaces in practice (e.g. animation tweening, image cross-fades, or Mixup data augmentation in machine learning).

9 Use NumPy to verify that for any three non-collinear vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$, their affine combinations indeed span a 2-D plane.

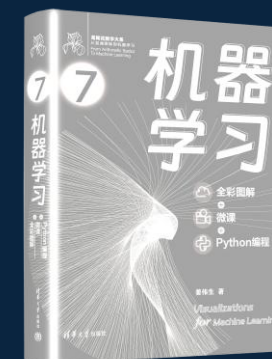
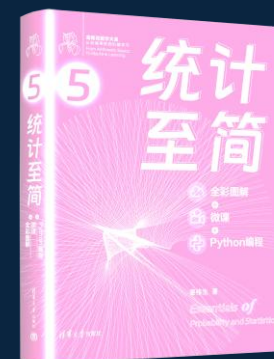
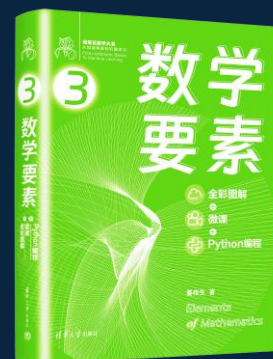
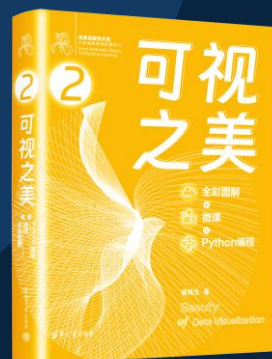
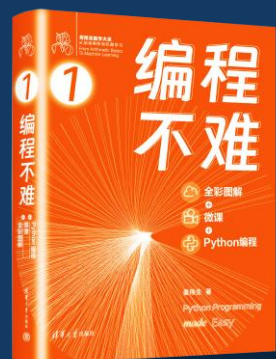
10 Design a Python exercise: given two images, implement a cross-fade animation using $k \cdot \text{img1} + (1-k) \cdot \text{img2}$ (linear interpolation of images).

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

